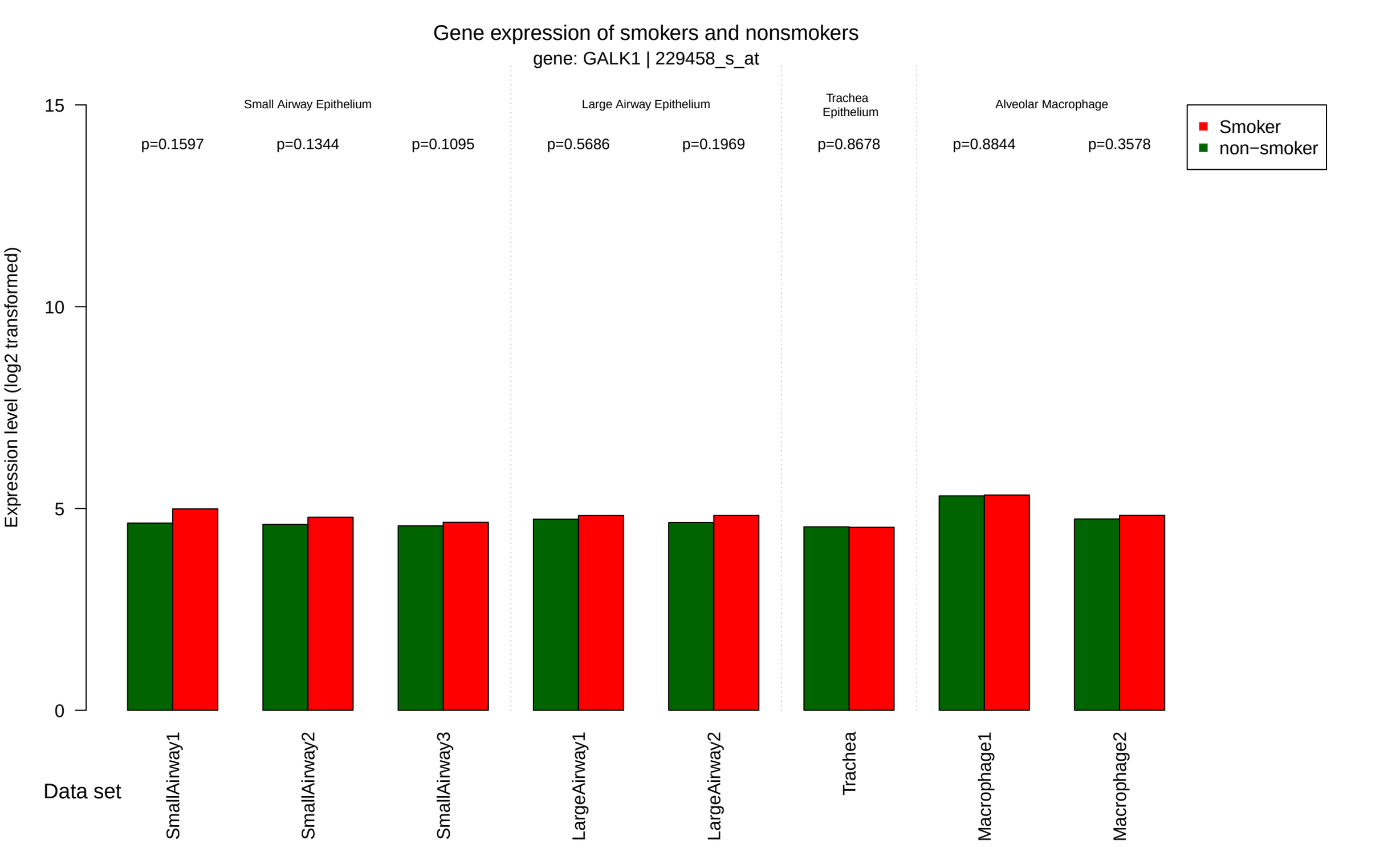
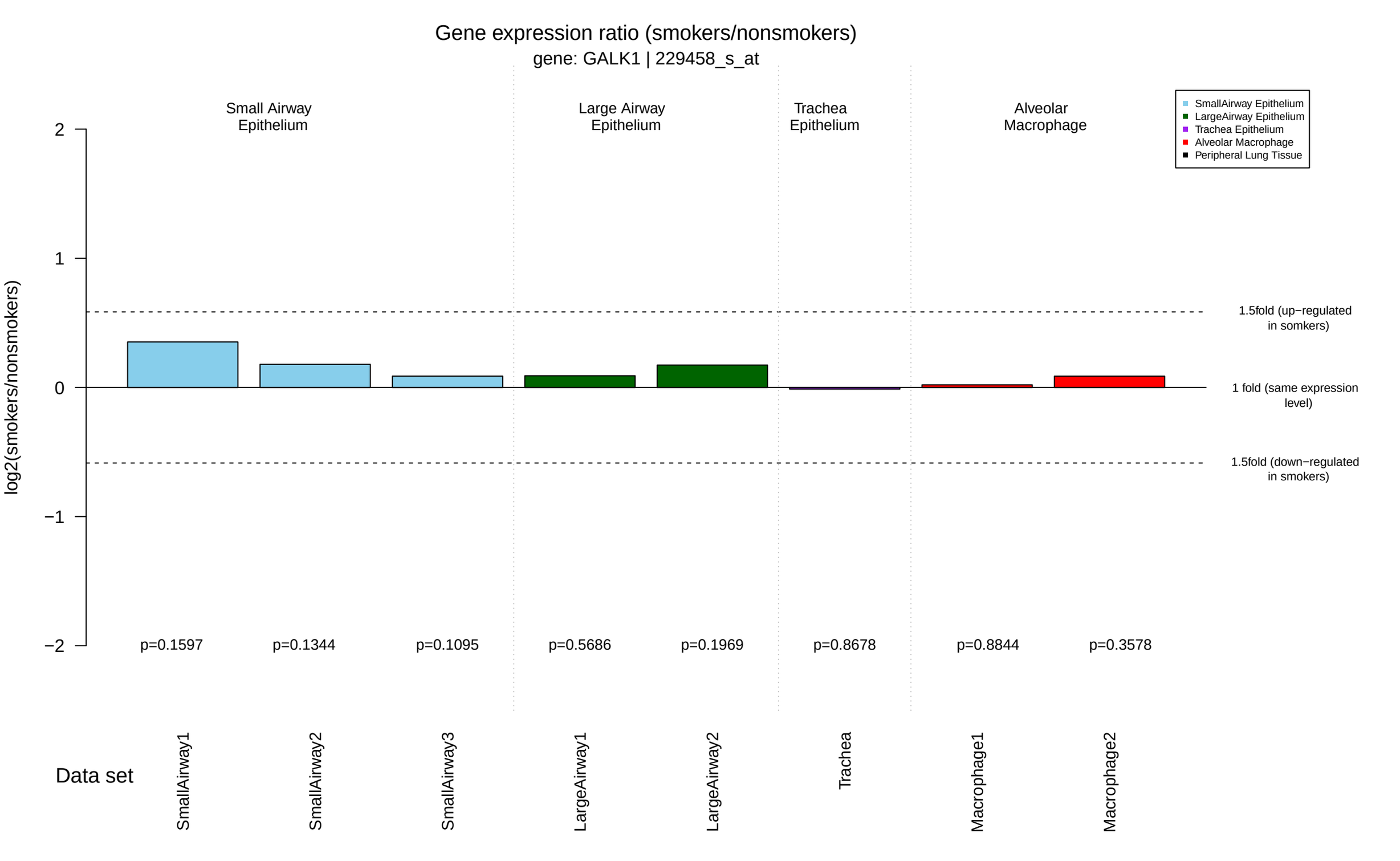
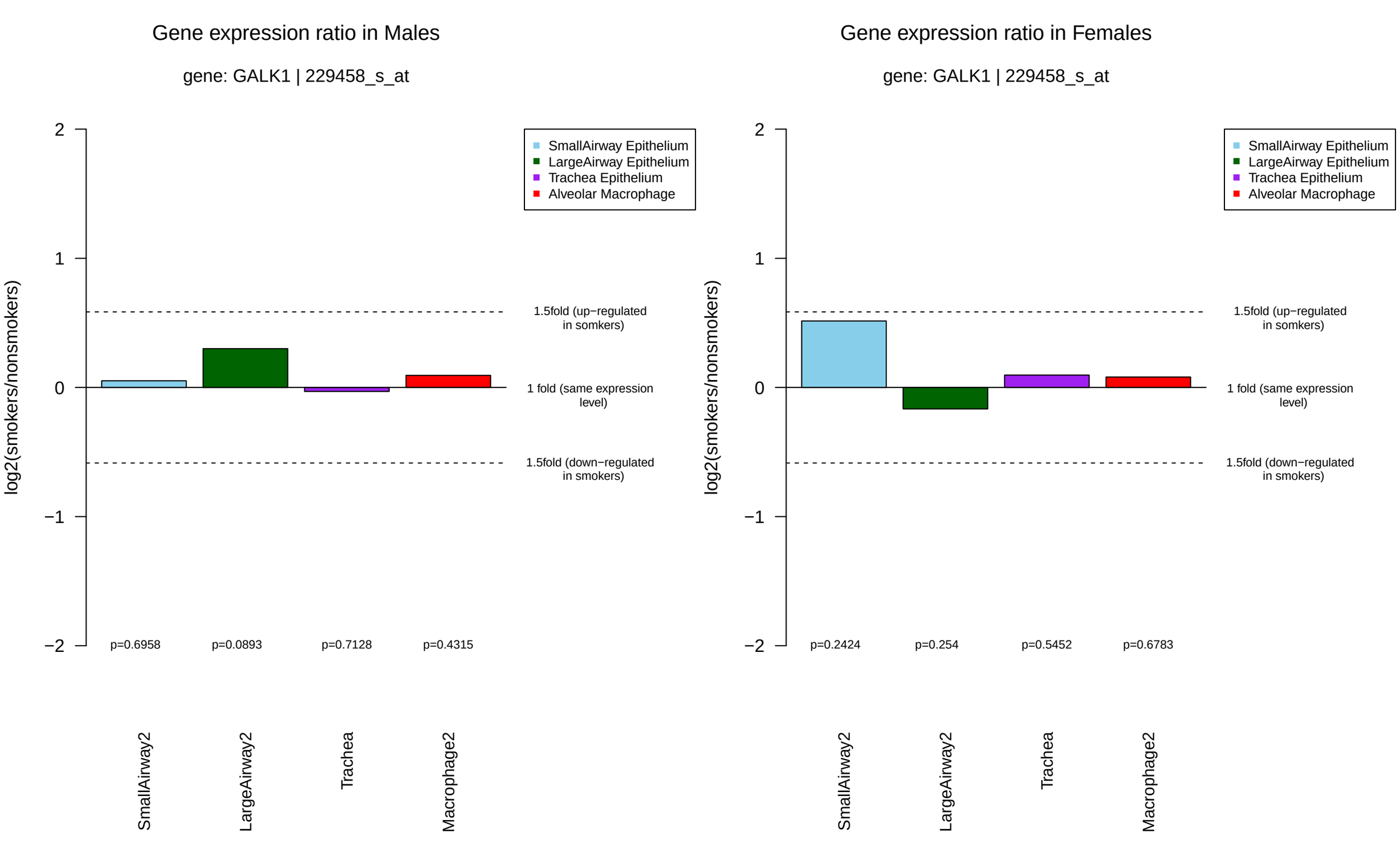
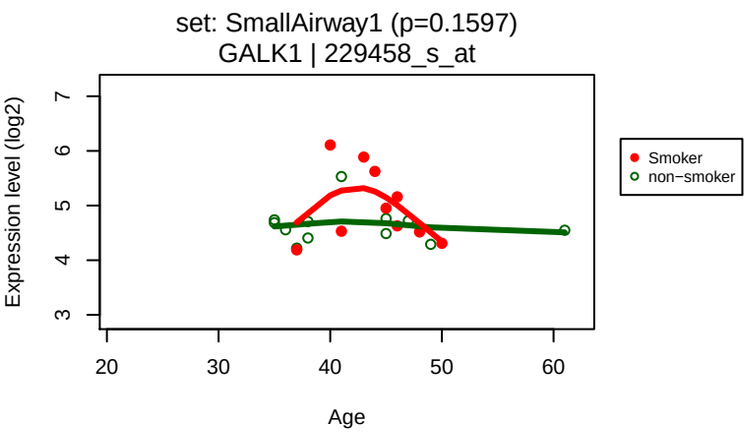
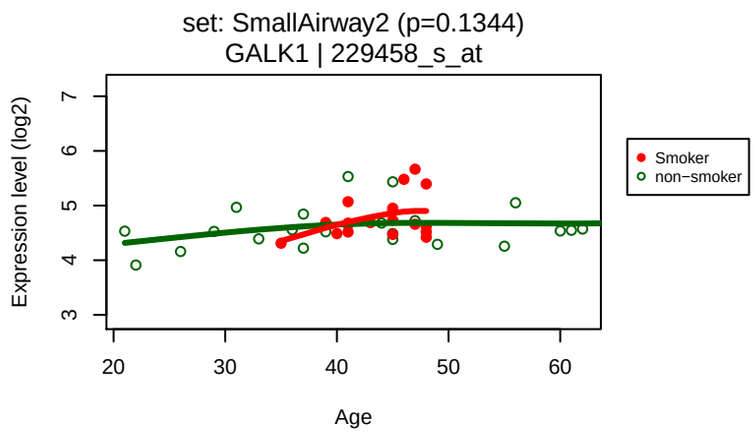
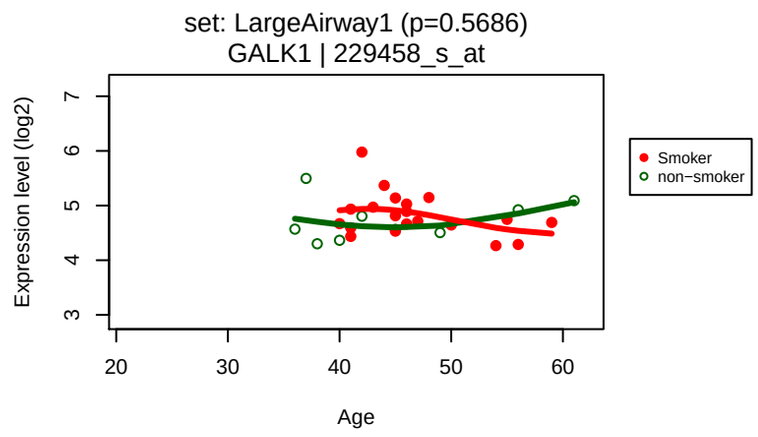

Gene expression of smokers and nonsmokers
gene: GALK1 | 229458_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1597
p=0.1344
p=0.1095
p=0.5686
p=0.1969
p=0.8678
p=0.8844
p=0.3578
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

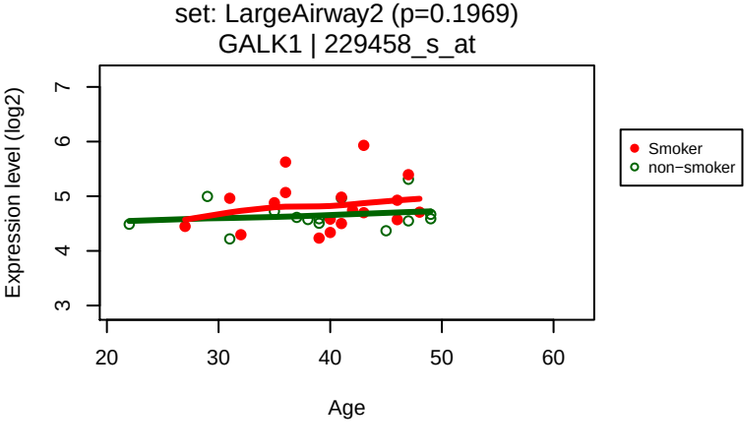

Gene expression ratio (smokers/nonsmokers)
gene: GALK1 | 229458_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1597
p=0.1344
p=0.1095
p=0.5686
p=0.1969
p=0.8678
p=0.8844
p=0.3578
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

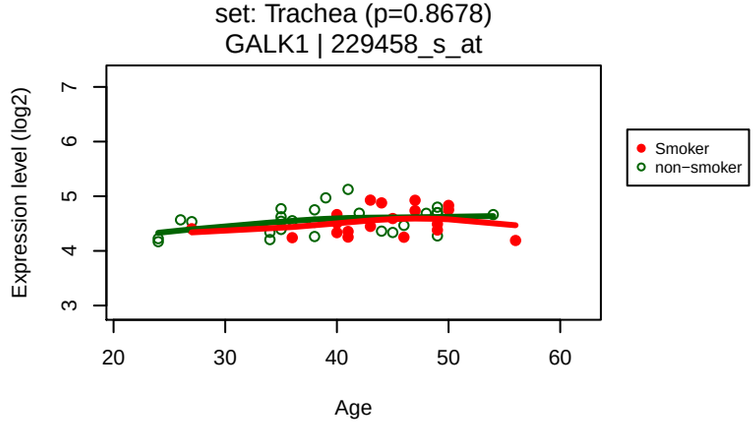

Gene expression ratio in Males
gene: GALK1 | 229458_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6958
p=0.0893
p=0.7128
p=0.4315
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GALK1 | 229458_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2424
p=0.254
p=0.5452
p=0.6783
SmallAirway2
LargeAirway2
Trachea
Macrophage2

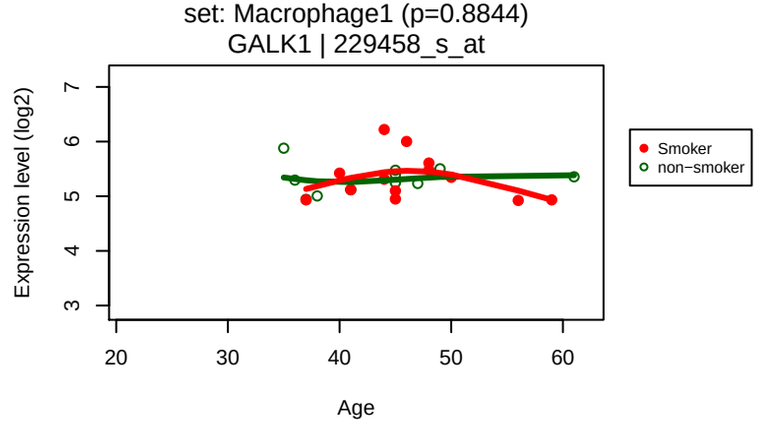

set: SmallAirway1 (p=0.1597)
GALK1 | 229458_s_at
Expression level (log2)
Age
Smoker
non−smoker

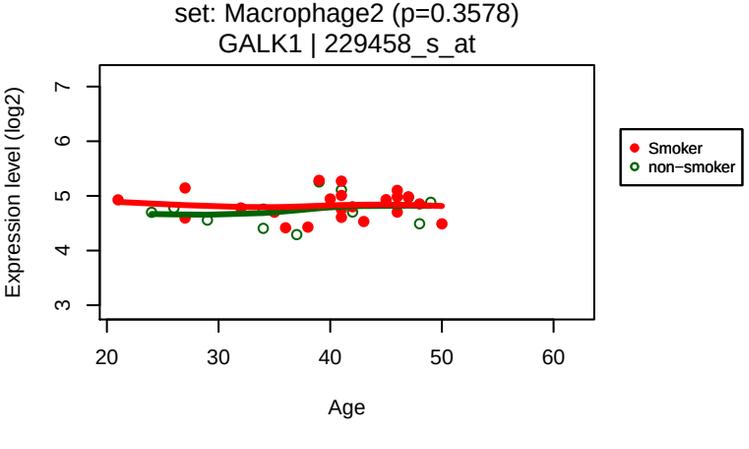

set: SmallAirway2 (p=0.1344)
GALK1 | 229458_s_at
Expression level (log2)
Age
Smoker
non−smoker

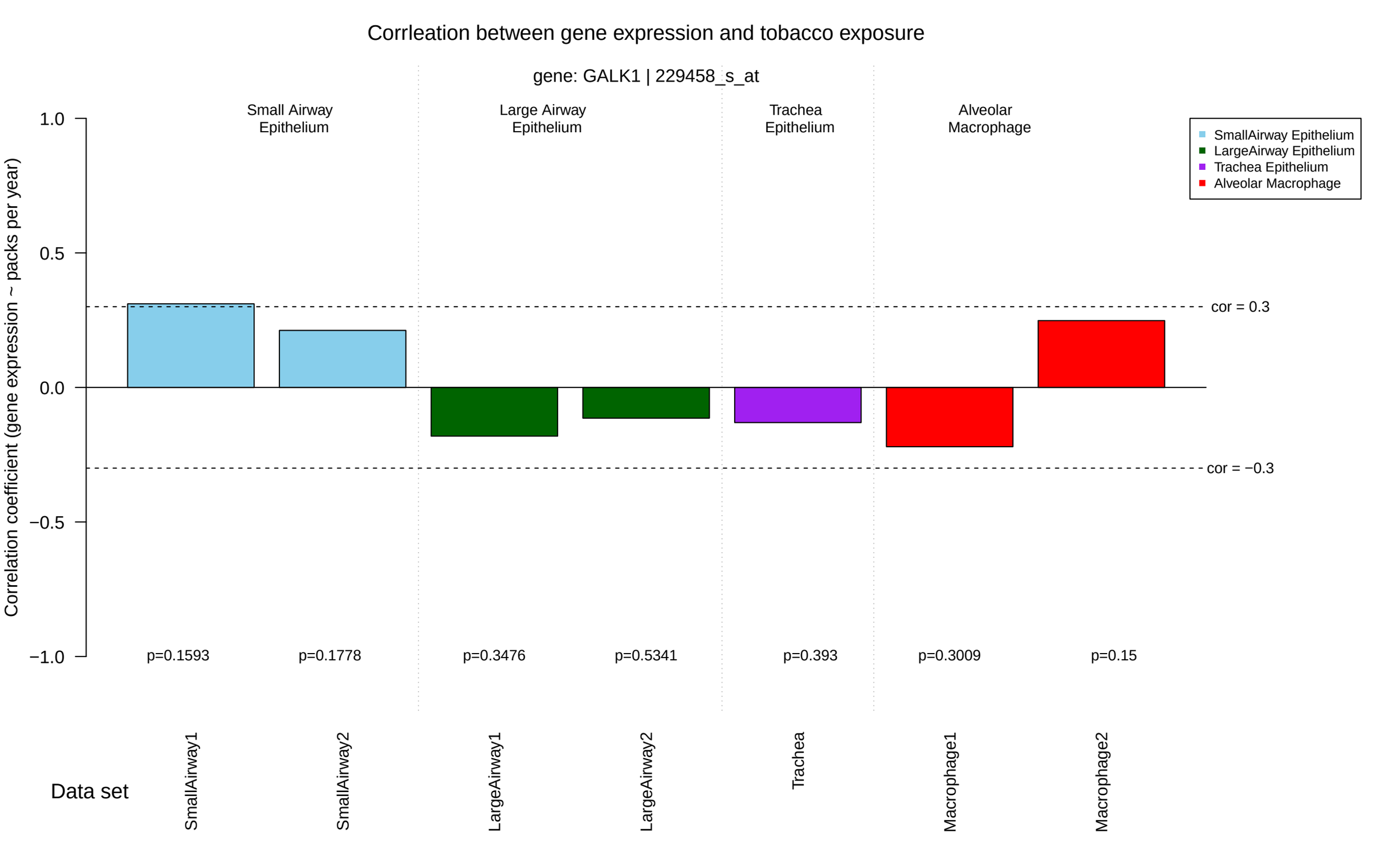

set: LargeAirway1 (p=0.5686)
GALK1 | 229458_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1969)
GALK1 | 229458_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8678)
GALK1 | 229458_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8844)
GALK1 | 229458_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3578)
GALK1 | 229458_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GALK1 | 229458_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1593
p=0.1778
p=0.3476
p=0.5341
p=0.393
p=0.3009
p=0.15
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2